

Segmenting Egocentric Work Videos into Described Sub-Tasks Without Labels

A geometry-first pipeline: dead-space removal, camera-pose transition detection, and grounded vision-language description

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September 2026

Abstract

We describe a pipeline that turns a long egocentric recording of manual work into a sequence of sub-task clips, each with a grounded natural-language description and a per-video summary, without training a segmentation model and without labels. The pipeline has four stages. (1) Ingest normalises every recording to 720p/15 fps by presentation timestamp. (2) A dead-space stage removes stretches in which nothing is visible or nothing happens, using four complementary detectors whose union is cut out and the remainder stitched; videos with ≤ 5 s of footage left are discarded. (3) A transition stage recovers the camera’s metric trajectory with Depth Anything 3 and places a boundary wherever the wearer was still, turned, walked a distance scaled to the space, and stayed—a *station change*. (4) The stitched video is cut at those boundaries and each clip is described by a 27 B vision-language model under strict grounding rules, with timestamped actions, a place guess, worded as a guess and supported by visible cues, and an explicit list of what could not be determined; a text-only pass over the ordered descriptions yields the video’s global context. On a corpus of 20 videos (3.4 h) the pipeline discards 3 videos, removes 1,621 s of dead footage from the rest, proposes 37 station changes on 17 videos in either of two pose settings (two DA3 models, or one metric model for both turns and trajectory; 35 station changes shared), and produces 54 clip descriptions and 17 global contexts per setting with no failed requests, in 54 to 57 minutes of total run time on one GPU after download. The cut is placed at the walk onset after the turn, read from the trajectory. Every boundary is explainable by three numbers, every parameter has a stated reason, and each stage records the alternatives that were tried and why they were rejected. We argue that this split of roles—deep models for representation, classical rules for decisions—is the right architecture when labels are absent and the footage spans kinds of video as different as a static bench camera and a continuously moving body camera. Appendix B unpacks the one-line rules of the paper with worked numbers.

Setting. 20 egocentric videos (10× 1080p/30 fps ten-minute body- or head-camera recordings of manual work; 10× 720p mobile clips of 15–60 s), 3.4 h in total, with no ground-truth boundaries or labels. One A800 80 GB GPU with 112 CPU cores.

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1 Introduction

Robot-learning pipelines that ingest human videos need those videos cut into pieces that each correspond to one goal: sub-tasks a policy can be conditioned on and a language model can name. Published systems either segment on reconstructed hand motion (VITRA [1]: the low points of wrist speed in world coordinates) or treat segmentation as a minor preprocessing step (PhysBrain [2]). Both assume visible, reconstructible hands of a single actor and a downstream captioner that merges pieces cut too finely. Our footage is harder in ordinary ways: wide-angle body cameras, hands often out of frame, long parked-camera stretches, a second person at work, and no labels of any kind.

Three constraints shaped the design.

- **No labels.** The boundary decision cannot be learned; it has to be a rule whose parameters can be tuned from reports and frame strips alone.
- **Two kinds of video in one corpus.** Some videos are a person at one bench for ten minutes with a few short trips away; others are a person who never stops moving (cleaning a hotel room). A rule that fires on motion fires all the time on the second; a rule that needs stillness never fires on the first.
- **The VLM is the costly component and the one most likely to believe what it should not.** It should see only footage that carries content, only in units that correspond to one place and activity, and it should be prevented from asserting what it infers.

The pipeline follows the same pattern VITRA uses: deep models supply the representation (camera pose from Depth Anything 3 [3], language from Qwen3.8 [4, 5]), classical rules make the decisions (what is dead, where a station changes, how a clip is cut). Figure 1 gives the overview. Section 2 describes the four stages and, for each, the alternatives that were run and why they lost. Section 3 gives results, Section 4 argues why an end-to-end model would have been the wrong choice here. Appendix A collects the parameters; Appendix B answers, with worked examples, the questions a reader is likely to have about the one-line rules.

2 Method

2.1 Stage 1 — Ingest and normalisation

Originals are fetched from object storage and re-encoded with FFmpeg [22] to 1280×720 at 15 fps (libx264 preset fast, 770 kbps target, 900 kbps cap, AAC 128 kbps). Two details matter. Frame-rate conversion uses ffmpeg’s `fps=` filter, which resamples by presentation timestamp; 90.9% of the corpus carries a wrong declared frame rate, and the usual `-r` option would drop about a sixth of the frames without warning. And nothing large ever crosses the link between the workstation and the GPU box, which runs at ~ 10 KB/s: originals are pulled onto the GPU box directly at ~ 10 MB/s per stream, and only JSON and markdown come back.

The re-encode costs nothing downstream. Camera poses computed on the 720p re-encode and on the 1080p original differ by a median of 0.29 m in position and 2° in heading over a ten-minute video, well inside the tolerances of every later rule.

Alternatives. Working at native resolution quadruples decode cost for identical poses. Copying videos over the link was attempted twice (full files, then 2 fps proxies) and abandoned. Cutting clips with ffmpeg’s segment muxer, used in an earlier experiment, lands on keyframes and skipped the first boundary without warning of every `.mov`; every clip is now an accurate cut with re-encode whose duration is probed against the intended one.

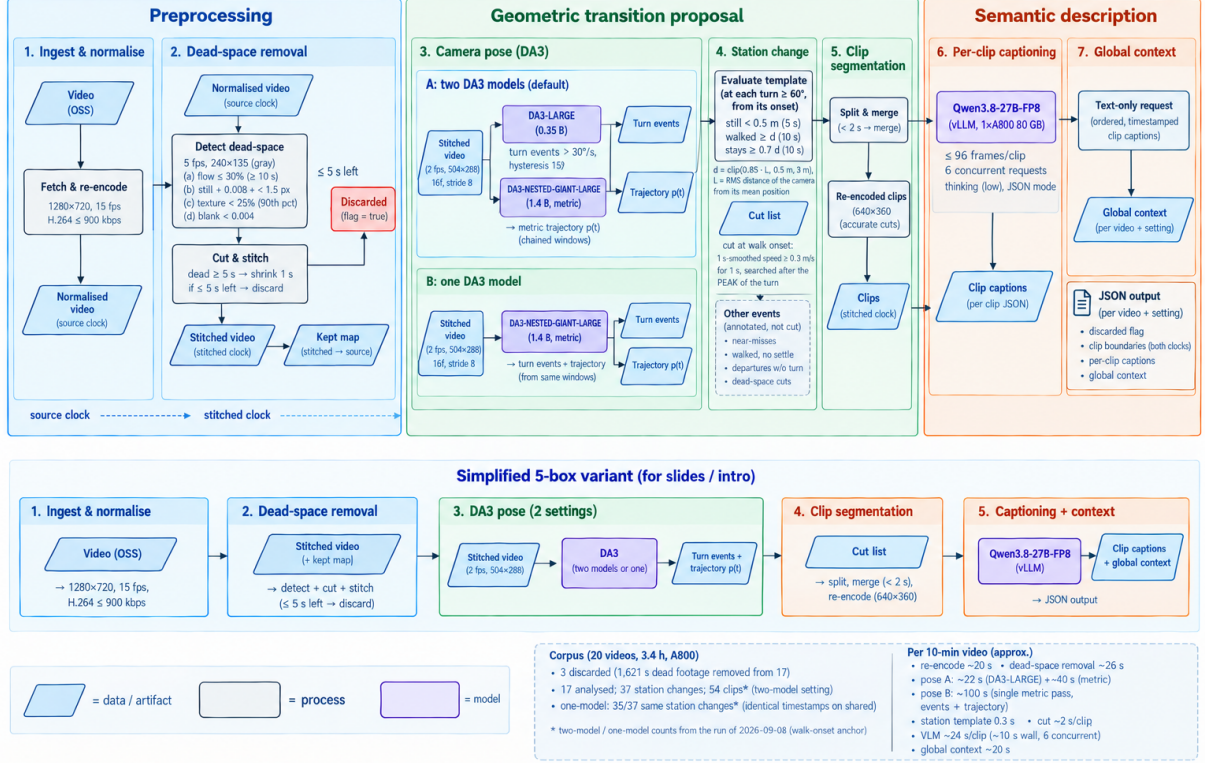


Figure 1: The pipeline. Left: preprocessing (ingest/normalise, dead-space removal with its four detectors, the kept-map between original-video and stitched-video time, the ≤ 5 s discard). Middle: geometric transition proposal (DA3 turn events and metric trajectory in the two pose settings, two models or one; the station-change template with the cut at the walk onset; clip segmentation). Right: semantic description (per-clip Qwen3.8 captions, global context, output JSON). The bottom row is the same pipeline collapsed to five boxes, with the corpus and per-video numbers.

2.2 Stage 2 — Dead-space removal

Egocentric recordings contain minutes of nothing: the camera parked on a surface, the lens against fabric, a black screen, a wall. Sending those to a pose model or a VLM wastes compute and produces confident descriptions of nothing. The stage runs on a 5 fps, 240×135 grayscale stream (OpenCV [20]) and combines four frame-level detectors (Table 1).

The union of the four is extracted as runs ≥ 5 s, shrunk by 1 s at each end so no cut eats the start of an action, removed, and the remainder stitched into one video (see App. B.3). A kept-map records the correspondence between stitched-video time and original-video time; every later timestamp carries both. If ≤ 5 s remain the video is discarded and recorded as such.

Why four detectors. Each was added because the previous set missed a case in the corpus. A relative rule compares a video with itself, so it finds dead stretches inside an active video but reads a video that is entirely a wall as active; the absolute-floor still detector fixes that but is blind wherever nothing can be aligned, such as a blank surface; the texture gate, compared against the 90th percentile rather than the median (a half-blank video has its median inside the blank zone), fixes that; and a black screen defeated all three, because relative rules compare black with black and the still detector returns no result when ORB finds no features. The absolute texture floor sits $4\times$ below the darkest normal video and $8\times$ above the black screen. Luminance was considered as the last gate and rejected: a covered lens, a white ceiling or fog have ordinary brightness and no texture.

Table 1: The four dead-space detectors. Relative rules compare a video with itself; absolute rules use floors set at the sensor-noise level of the analysis stream. The still and flow rules are unpacked in Appendix B.

detector	rule	type	what it catches
optical flow	Farnebäck [11] mean flow $\leq 30\%$ of the video’s median over a segment ≥ 10 s (see App. B.2)	relative	the parked camera inside an otherwise active video
still	ORB [12] + MAGSAC [13] homography per frame pair; compensated residual < 0.008 and camera motion < 1.5 px over a 3 s window; a 30 s long-lag alignment residual < 0.02 covers hand shake (see App. B.1)	absolute	videos that are quiet throughout
texture	mean image gradient $< 25\%$ of the video’s own 90th percentile for ≥ 5 s	relative, against the 90th percentile	lens on a blank surface, where flow reads noise and there is nothing to align
blank	mean image gradient < 0.004 for ≥ 5 s; blank frames also vote in the still detector with camera motion taken as zero	absolute	nothing in the frame at all: black or blown-out screen, covered lens

Alternatives. A plain motion threshold, or ffmpeg’s `freezedetect`, keeps the commonest dead footage there is—the camera swinging while the wearer walks or repositions it—and cuts the pauses between actions, which are boundaries, not dead footage. An earlier quantile-based detector, built on a hand-written Harris [14] + NCC [15] + RANSAC [16] homography, switched on and off repeatedly inside dead regions because the quantiles were set by the dead region itself (88 s caught of ~ 240 s, in five fragments), lost a third of the frame pairs on the fast-moving video and made the residual *worse* than raw on the static one (App. B.4 and B.5). Per-frame thresholding without run extraction cuts the video into hundreds of tiny pieces. More expensive checks (whether image features stop changing, a VLM call per candidate run, audio energy) were not needed: the four cheap detectors handled all 20 videos.

Corpus result. 3 videos discarded (two were 2 s of padding after a fully dead recording; one a black screen), 1,621 s removed from the other 17, 0 s removed from the two continuously active videos. About 26 s per ten-minute video.

2.3 Stage 3 — Transition points from camera pose

The boundary we want is a change of workplace: the person finishes at one station, turns, walks, and starts at another (see App. B.6). Detecting it needs the camera’s position over time, not just its image motion.

Pose. Depth Anything 3 (DA3) [3] runs on the stitched video at 2 fps and 504×288 in 16-frame windows with stride 8, so each frame is seen twice and the estimate from the window where it is more central is kept. Two quantities are needed: per-frame relative rotation, from which turn events are extracted (angular speed above $30^\circ/\text{s}$ with hysteresis to 15, swept angle, net yaw and pitch), and the camera trajectory $p(t)$ in metres, obtained by chaining the per-window poses through the shared frames. Only the metric variant, DA3-NESTED-GIANT-LARGE (1.4 B), has a scale that is consistent across windows, and a metric trajectory is what makes a displacement threshold meaningful. The pipeline is run in **two pose settings**, kept side by side throughout the results: *two DA3 models*, where DA3-LARGE (0.35 B) supplies the turn events and the metric model the trajectory, and *one DA3 model*, where a single pass of the metric model supplies both from the same windows (see App. B.7).

The station-change template. At every turn onset t_0 with swept angle $\geq 60^\circ$, three conditions are tested on $p(t)$ (Table 2).

Table 2: The station-change template. L is the root-mean-square distance of the camera from its average position over the whole trajectory (the size of the space used), so d scales with the space the person used. Worked numbers in App. B.8.

condition	test	value
still before	90th-percentile distance from the mean position over $[t_0 - 5\text{ s}, t_0]$	$< 0.5\text{ m}$
walked	maximum distance from that position within $[t_0, t_0 + 10\text{ s}]$	$\geq d$
stays	fraction of samples $\geq 0.7d$ from the old place in the 10 s after arrival	≥ 0.8
d	$\text{clip}(k \cdot L, 0.5\text{ m}, 3\text{ m})$	$k = 0.85$

k was 1.0 until a reviewed departure (mov-04 at 0:29, a 1.42 m walk against $d = 1.57\text{ m}$) was judged a true station change; at $k = 0.9$ it still missed by 0.2 mm, at 0.85 it is a cut and two other videos gain one cut each (mov-08 at 1:01.5, mov-09 at 1:24.5).

The turn only sets the timing: it says *when* to test. Displacement and the two stay conditions make the decision. The cut itself is placed at the **walk onset**: the first moment after the peak angular speed of the turn where the 1 s-smoothed camera speed stays $\geq 0.3\text{ m/s}$ for two consecutive samples, searched up to 10 s after t_0 (if no such pair exists, the first single sample $\geq 0.3\text{ m/s}$; if none, t_0). That is the instant the person leaves the old station, read from the trajectory rather than from the turn model; on this corpus the two pose settings place 27 of their 35 shared cuts at the same sample (see App. B.10). The peak of the turn was the anchor in the first runs and is still reported next to every cut. The still-before condition removes turns made mid-walk; the stays condition removes trips where the person goes to get something and comes straight back; d scales with the space the person used, because a station change is 0.7 m in a kitchenette and 5 m in a hall, and the clamps stop a static video from driving d to zero or a huge hall from driving it to infinity (see App. B.8). The time windows are fixed values chosen from how people work and are not adapted.

Beyond the cuts, the stage reports what it rejected and why: walks that fell just short of d (0.8–1.0 d), turns followed by a walk that failed a stay condition, departures that fit the template without a qualifying turn, and the dead-space cuts as pause boundaries. These lists are what a reviewer tunes from.

Why this rule and not the others. Fourteen alternatives were run on this corpus before the template was fixed; the important ones are these.

- *Shot-boundary detection* [19] is ruled out for continuous takes: it detects cuts between shots, of which there are none, while head turns and auto-exposure produce the largest deltas.
- *Low points of the optical-flow motion curve* (VITRA’s cut rule [1] applied to image motion)

produce very short actions, 120–360 per ten minutes, median ~ 2 s. Smoothing harder to reach sub-task scale destroys the evidence: at $\sigma = 1\text{--}2$ s only 35–66 % of boundaries survive a $\times 2$ change of σ . Ranking the low points by how deep they are [21] restores stability (0.84) but not meaning: on egocentric footage the low points are pauses in *head* motion, and on a featureless stretch the flat-zero trace returns one 283 s “segment”.

- *Homography-compensated residual motion* [17] removes 15–25 % of pixel change as ego-motion when the camera moves moderately, makes the signal 36–95 % worse on a static camera, fails outright on fast camera motion (inlier ratio ~ 0.5 , residual above raw), and moves few boundaries (agreement with plain differencing F1 0.84–0.91). App. B.5 gives the details.
- *Appearance clustering* (TW-FINCH [27] on a classical descriptor) yields contiguous 20–30 s stretches of similar appearance but without meaning: two actions at the same bench look identical and a lighting change looks like a new action; it disagrees with the motion methods at F1 0.00–0.23, and without ground truth two families that never coincide cannot validate each other.
- *A classical camera-pose pipeline* (ORB + homography/essential matrix [17] at 10 fps) matched DA3 on turn timing but gave no magnitudes, produced nothing on blank or blurred frames (48 % of mov-01 unalignable), and could not recover the intrinsics.
- *Rotation alone* fires on every glance: 46 sweeps before the first real move on one video, 121 turns on another.
- *Translation-rate thresholds* need a video-relative floor that is undefined on a static video and meaningless on a continuous one, and rate integrates to path length, not displacement: mopping produced 214 m of path inside a 4 m room.
- “*A translation run starts after a turn*” never fires when the camera never stops. *Otsu splits* [18] of the rate distribution found no bimodality; the splits landed on noise floors. *Scaling d by the area the person covered* was rejected because two videos covering the same area behaved in opposite ways. *Chaining the non-metric trajectory* reported a 3.5 m walk as 35 units.
- *Hand-pose speed*, the VITRA signal, was not run: hands are out of frame for long stretches, one video has a passive wearer with a second person working, and per-hand segmentation has no meaning across actors.

Corpus result. 37 station changes on 17 videos in each setting, 35 of them the same departure in both; 11 videos correctly have none (static camera, or a single workplace). The pose stage costs ~ 1 min per ten-minute video on the GPU with two models and ~ 65 s with one; the template runs in 0.3 s on the CPU.

2.4 Stage 4 — Clips and grounded description

Clips. The stitched video is cut at $[0, T_1], [T_1, T_2], \dots, [T_n, \text{end}]$; a boundary that would leave a clip under 2 s is merged into its neighbour. Clips are accurate cuts with re-encode (640×360 , source frame rate, no audio). The corpus gives 54 clips per setting, of 3.5–517 s (two models) and 6.1–517 s (one model).

Serving. Qwen3.8-27B-FP8 [4, 5], a vision-language model with video input, served by vLLM 0.28 [6] on the A800. Four settings were forced by the hardware and the model family: the Ampere card has no native FP8, so the weights run through Marlin [7] weight-only kernels at ~ 45 tokens/s per stream; the model is a hybrid of full-attention and gated linear-recurrence layers [10]

whose recurrent state vLLM keeps in a Mamba-style cache [9], so concurrency is capped at 32 sequences or startup aborts; FlashInfer’s [8] sampler is disabled because it JIT-compiles at startup and needs a toolchain the box lacks; and frame sampling is controlled per request through the media loader’s `fps` (the processor-level parameter the model card names is ignored in this version). Each clip is sampled at `fps = clip(96/duration, 0.2, 2.0)`, i.e. at most ~ 96 frames and ~ 11 k prompt tokens. Six requests run concurrently: 572s for 50 clips against 1,215s sequentially.

Prompting. Thinking is on at low effort, with the model card’s thinking-mode sampling (temperature 1.0, top- p 0.95, top- k 20, fixed seed) and a 4,000-token budget. The answer is requested in JSON mode and validated in code against a fixed schema; a fallback with thinking off and a schema grammar (see App. B.9) exists and was never triggered. The schema grammar could not be combined with thinking directly: it leaves whitespace free and the model fills the answer with it until the token limit, reproducibly. The system prompt states the grounding rules: describe only what is visible; do not infer identity, emotion, intention or off-screen events; name places and objects by what is seen and mark inferred identities as guesses; write N/A for a string that cannot be determined and leave lists empty rather than filling them with N/A. The per-clip record has eleven fields (Table 3).

Table 3: The per-clip record.

field	content	purpose
<code>setting_guess,</code> <code>setting_evidence[]</code>	one place type, worded as a guess (“seems like a hotel bedroom”) and the visible cues it rests on; the validator moves any additional alternative into <code>uncertain_about</code>	separates the inference from its evidence; keeps the guess checkable
<code>camera</code>	viewpoint and camera motion	keeps panning out of the action list
<code>objects[]</code>	visible description first, “likely X (cue)” when the identity is inferred	the same guess wording at object level
<code>actions[]</code> of <code>{t, action}</code>	what the wearer does, verb + object, with t = seconds from the clip start read from the frame timestamps	turns a description into within-clip boundaries
<code>others_actions[]</code>	the same for other visible people	a passive wearer with someone else working no longer yields an empty clip
<code>hands</code>	which hands are visible and what they hold	
<code>text_or_labels_visible[]</code>	legible text only, quoted	the main hallucination channel; reported for review, not trusted
<code>one_line_summary</code> <code>visibility</code>	one sentence clear / partly_obscured / mostly_unclear, by a rubric on how much of the clip shows hands and workpiece in focus	replaces a free-form confidence that came back “low” on 44 of 50 clips
<code>uncertain_about[]</code>	what could not be determined, including alternative readings of the place	doubts made explicit instead of forced into a guess

Global context. A text-only request per video receives the ordered clip descriptions with every action, the wearer’s and others’, carrying its time on the video’s timeline, and returns an

overall activity worded as a guess, a location guess with cues, the sub-task sequence as `{start, subtask}` where starts are taken from the action timestamps (a sub-task may begin mid-clip and continue across clips), the people visible, and open uncertainties.

How the prompt evolved. Three versions were run on the full corpus. v1 asked for a free confidence, stated the setting as fact and gave untimed actions; confidence came back “low” on 44 of 50 clips, settings were asserted from cues (“café counter” for a hotel kettle tray, “boat interior” for a workshop), and actions could not be placed in time. v2 introduced the guess-worded setting with evidence, timestamped actions and the visibility rubric, but let several alternatives be listed inside the guess (“workshop, hangar, or ship structure”), left clips with a passive wearer empty, and its global sub-task list simply copied the clip boundaries. v3 restricts the guess to one place type, adds `others_actions`, and instructs the global step to cut sub-tasks at action timestamps. An independent OCR cross-check of the visible-text field was built, confirmed 9 of 35 claims, and was then removed by decision so that the pipeline uses the VLM only; the field is reported unverified.

Alternatives. Asking the VLM for boundaries on the long video was ruled out: temporal localisation is the documented weak spot of these models, and a ten-minute video at 1 fps is ~ 150 k visual tokens per call against 11 k per clip here. Fixed-interval chunks cut through the middle of actions and misalign captions (VITRA’s ablation puts them below motion-based cuts). Over-segmenting and merging in language space, the step VITRA depends on, is unnecessary when the transition stage emits 37 cuts rather than hundreds; within-clip structure comes from the timestamped actions instead.

3 Results

The two settings describe the same footage in nearly the same words: their clip boundaries agree on 29 of 37 cuts within 1 s, and where they differ the captions differ only in which clip a given action falls into. The caption-level counts in Table 4 move by a few units between settings, which is the sampling variance of thinking-mode decoding as much as the cuts.

Two videos can be checked against what is known of their content. mov-04, a hotel room being cleaned, comes out as “seems like a hotel or guesthouse room” with cues (bed with white duvet, framed guest notice and wifi card, service counter with a tray of cups, tiled bathroom with a glass shower screen), and its sub-tasks run: 0:00 tidying the tea tray, 0:32 rinsing the pot at the sink, 1:06 wiping the counter and chair, 1:57 wiping the desk and dresser, 2:08 clearing the drawer, 2:27 the bed edge and bedside, 3:24 the washbasin and mirror, 3:35 the wet bathroom floor, 4:08 squeegeeing the glass panel, 7:19 the tiled wall and fixture, 7:46 wiping and mopping through to the bedroom—with the 1:06, 2:08 and 3:24 boundaries inside clips. mov-01, a worker in a hi-vis vest on a construction floor, comes out as a site inside a dome-shaped hall with scaffolding, where a red power unit is dragged, repositioned and connected, followed by a short break and a walk outdoors to carry a wrapped object; the one-model setting, with the same clips, reads the same sequence. mp4-05, a black screen, is discarded before the VLM sees it.

4 Why not an end-to-end model

The question deserves a direct answer, because a single video-language model that emits segments and captions is the obvious alternative.

Labels. Supervised temporal action segmentation (MS-TCN [23], ASFormer [24], DiffAct [25]) needs dense frame labels that do not exist here. Unsupervised methods (ABD [26], TW-FINCH [27], generic event boundary detection [28]) are benchmarked on third-person cooking

Table 4: Corpus results (20 videos, 3.4 h, one A800), both pose settings, run of 2026-09-08 with walk-onset cuts and prompt v3. Dead-space removal and the discards are shared.

	two DA3 models	one DA3 model
videos	20 $\rightarrow$ 3 discarded after	same
dead footage removed	dead-space removal, 17 analysed	
	1,621 s from the 17 analysed	same
	videos	
station changes / clips	37 / 54	37 / 54 (35 station changes shared; 27 at the same sample)
description requests	54, 0 failed, 0 fell back to no-thinking; 586 k tokens	54, 0 failed; 594 k tokens
wearer actions with timestamps	333, all inside their clip	329
other people’s actions	48	54
setting guesses	54 single place types; additional alternatives moved to uncertain_about	54
visible-text claims	35, reported as read, unverified	48
visibility	clear 24 · partly obscured 24 · mostly unclear 6	clear 20 · partly obscured 30 · mostly unclear 4
global sub-tasks	97 across 17 videos, 53 starting more than 2 s from any clip boundary	94, 52 mid-clip
total run time, one A800	re-encode + dead space 15 min (shared), pose 20 min, clips 2 min, captions 10 min and global contexts 9 min wall with 6 concurrent requests; $\approx$ 57 min after download	pose 18 min (one pass, 65 s per video), otherwise the same; $\approx$ 54 min

footage; the one we ran, TW-FINCH, produced stretches of similar appearance without meaning. Our decision layer has four parameters with physical meaning and was re-tuned three times in a day from reports and frame strips alone.

Explainability, and therefore debuggability. Every cut carries a turn angle, metres walked and stay times; every rejected turn carries the condition it failed. The three defects found during the run—a black screen passing as active, a cut placed 1.5 s early, a walk just short of the displacement threshold—were each located in one number and fixed in one line. An end-to-end model would have produced the same wrong boundaries with nothing to adjust.

The deep models are used where they are strong and nowhere else. DA3 supplies geometry a classical pipeline could not: metres, intrinsics, robustness on blank and blurred frames. Qwen supplies language. Neither decides where a segment starts. Both are used unchanged, publicly available and replaceable; the rule between them did not change when DA3 replaced the ORB pipeline and will not change if a better VLM replaces Qwen.

Cost scales with places visited, not with footage length. The VLM sees at most 96 frames per clip: 54 clips and 586 k tokens for 3.4 hours of video. The same corpus fed to a VLM at 1 fps would be tens of millions of visual tokens, spent on a model that is weak at exactly the localisation being asked of it. The dead-space stage removes 1,621 s the VLM never sees and discards three videos outright.

Different kinds of video are handled by stated rules. A static camera yields zero cuts, a continuously moving one yields place changes, a half-dead one is stripped first, an empty one is discarded. A single learned model would need to have seen all of these in training.

The limit is explicit. Sub-tasks that happen within a metre of each other—counter, shower, toilet in one bathroom—are invisible to any pose rule and are recovered only through the timestamped actions in the descriptions. If labels are ever produced for that level, a learned within-station segmenter is where they should be spent, and these descriptions are the obvious training data for it.

A Parameters at a glance

stage	parameter	value	set by
1	resolution, frame rate, bitrate	1280×720 , 15 fps (timestamp resample), 770/900 kbps	downstream needs; corpus metadata errors
2	analysis stream	5 fps, 240×135 grey	minute-scale phenomena
2	flow rule	$\leq 30\%$ of median, segments ≥ 10 s (App. B.2)	relative; parked camera inside active video
2	still floors	residual 0.008, camera 1.5 px, long-lag 0.02 (App. B.1)	absolute; sensor-noise level at this resolution
2	texture gate	$< 25\%$ of p90, ≥ 5 s	relative, against the 90th percentile
2	blank floor	gradient < 0.004 , ≥ 5 s	absolute; $4\times$ below darkest normal video
2	run extraction	≥ 5 s, 1 s pad; discard if ≤ 5 s left (App. B.3)	keep action onsets; nothing to analyse below 5 s
3	DA3 sampling	2 fps, 504×288 , 16-frame windows, stride 8	cost vs turn timing
3	turn gate	swept $\geq 60^\circ$	timing only
3	still / walked / stays	5 s & 0.5 m / 10 s & d / 10 s & $0.7d$ at ≥ 0.8 (App. B.6, B.8)	human timescales; net displacement
3	d	clip($0.85 \times L$, 0.5 m, 3 m), L = RMS distance of the camera from its average position (App. B.8)	scales with the space; clamps protect videos where the camera barely moves or moves far in one go
3	cut position	walk onset: 1 s-smoothed speed ≥ 0.3 m/s for 1 s after the peak of the turn (App. B.10)	the instant of leaving; read from the trajectory, independent of the turn model
3	walking speed floor	0.3 m/s over two 0.5 s samples	below normal walking, above pose jitter
4	clip minimum	2 s	DA3 sample rate
4	frames per clip	≤ 96 (fps = clip($96/\text{duration}$, 0.2, 2))	~ 11 k prompt tokens
4	decoding	thinking on, low effort; T 1.0, top- p 0.95, top- k 20, seed 0; 4,000 tokens	model card thinking-mode settings
4	output	JSON mode + schema validation; grammar fallback with thinking off	grammar + thinking fills the answer with whitespace
4	concurrency	6 requests; server capped at 32 sequences	recurrent-state cache limit

B You might wonder?

Short, example-based unpackings of rules that are stated in one line in Sections 2.2, 2.3 and 2.4.

B.1 What does “ORB + MAGSAC homography; compensated residual < 0.008 and camera motion < 1.5 px over a 3 s window; 30 s long-lag residual < 0.02 ” actually mean?

Take two consecutive frames of the 5 fps, 240×135 analysis stream, a fifth of a second apart.

ORB + MAGSAC homography. ORB [12] finds a few hundred corner-like points in each frame and matches them. MAGSAC [13] fits a homography, a single 3×3 transform that maps frame $t-1$ onto frame t , while ignoring points that do not agree with the majority. On a body camera that transform is the camera’s own motion, a head turn or sway.

Camera motion, in px. Push the four corners of the frame through the homography and measure how far they moved. Standing still with normal hand shake gives about 0.1 to 1 px per frame at this resolution; a head turn gives 5 to 30 px. The still detector requires under 1.5 px.

Compensated residual. Warp frame $t-1$ with the homography so it lies on top of frame t , then take the mean absolute difference of the two, on intensities scaled 0 to 1. What survives the warp is change the camera motion does not explain: hands moving, an object being lifted. Example: a head turn alone gives a large raw frame difference but a compensated residual near 0.002, because after alignment the two frames match; a hand wiping a table gives a residual around 0.02 to 0.08 no matter what the head does. The floor of 0.008 is about two grey levels out of 255, the level of sensor noise.

Over a 3 s window. A frame counts as still when both conditions hold in at least half of the 15 frames around it, so a single blurred or misfitted frame does not flip the decision.

30 s long-lag residual. Hand shake can push the frame-to-frame residual just above the floor in bursts even when nothing happens. So the same alignment is also done between frame t and the frame 30 s later. If the scene is unchanged over 30 s, that residual is small, under 0.02, and the frame is still called still. mp4-01, a camera pointing at a wall, went from 61 % still to 99 % with this check.

Blank frames get no ORB points at all, so the homography cannot be fitted; those frames vote with camera motion 0 and the raw frame difference, which is the fix added for the black screen.

B.2 What does “Farnebäck mean flow $\leq 30\%$ of the video’s median over a segment ≥ 10 s” mean, and what is a flow trace or a segment?

Same analysis stream, 5 fps at 240×135 .

Farnebäck flow. For each pair of consecutive frames, Farnebäck’s method [11] estimates a motion vector at every pixel, how far that pixel appears to have moved between the two frames. Averaging the vector lengths over the whole frame gives one number per pair, the mean flow in pixels. A hand wiping a table gives about 0.5 to 1 px, a head turn 5 to 30 px, a parked camera near 0.

Flow trace. The list of those per-pair numbers, one per frame step. At 5 fps a 60 s video gives 300 numbers:

```
t (s):    0.0  0.2  0.4  0.6  0.8  1.0  1.2  1.4  ...
flow px: 0.8  1.2  0.4  0.1  0.3  2.5  3.0  0.2  ...
```

Plotted over time it is a bumpy line: bumps where the camera or hands move, valleys where they pause.

The video’s median. The median of that trace is the video’s own typical amount of motion. On mov-01, half active workshop footage and half a camera lying on a blank surface, it is about 0.1 px per frame; on a cleaning video where the person never stops, about 0.3.

Segments. The trace is cut at its valleys. The line is lightly smoothed (Gaussian, $\sigma = 0.15$ s), every local minimum that is at least a small dip (a dip of at least $0.15\times$ the trace’s standard deviation relative to its neighbours [21]) is a cut point, and cuts closer than 1 s apart are dropped. The pieces between cuts are the segments. In the example above, the valleys at 0.6 s and 1.4 s give segments 0.0–0.6, 0.6–1.4, and so on. Each segment is one short burst of motion, typically 1 to 5 s, and its mean flow is its average height. This is the same low-point rule the short-action cut of VITRA [1] uses.

$\leq 30\%$ **of the median over ≥ 10 s.** Walk along the segments and find runs where every segment’s mean flow is under 30 % of the video median. If such a run covers at least 10 s, it is dead. A 3 s pause between two actions is one low segment, under 10 s, so it survives: pauses are boundaries, not dead footage. A camera parked for four minutes is dozens of low segments in a row, so the whole run goes. On mov-01 the blank tail reads 0.0003 px against a median of 0.10, and the entire 279 s tail comes out as one dead run.

Why segment first rather than threshold raw frames: raw flow is noisy frame to frame. One jittery frame inside a dead stretch would otherwise break it into pieces, and one still frame inside an action would look like a tiny dead spot.

Why relative. The rule adapts to the camera and to the person: it finds “much quieter than this video usually is”. That is also the case it cannot detect, and the reason the still detector (App. B.1) exists beside it: a video that is quiet from start to finish has a median that is itself the quiet level, so nothing falls under 30 % of it. mp4-01, the camera on a wall, read as fully active to this rule.

B.3 What does “the union of the four is extracted as runs ≥ 5 s, shrunk by 1 s at each end, removed, and the remainder stitched into one video” mean?

Each of the four detectors marks individual frames of the 5 fps stream as dead or not. This is how those per-frame marks become the cuts that are actually applied.

Union. A frame is dead if any one of the four detectors says so. One stretch of a video:

```
t (s):      100 101 102 103 104 105 106 107 108 109 110 111 112
flow:       .  .  D  D  D  D  D  D  D  .  .  .  .
still:      .  .  .  D  D  D  D  D  D  D  D  .  .
texture:    .  .  .  .  .  .  .  .  .  .  .  .  .
blank:      .  .  .  .  .  .  .  .  .  .  .  .  .
union:      .  .  D  D  D  D  D  D  D  D  D  .  .
```

Runs ≥ 5 s. Consecutive dead frames form a run. Here the run is 102 to 111, nine seconds, so it qualifies. A run shorter than 5 s is thrown away and its frames are kept: a 2 s dead patch is usually a pause inside an action, and cutting it would only make the video jump.

Shrunk by 1 s at each end. The run’s edges are pulled inward by one second, so 102–111 becomes 103–110. The detectors decide with windows of a few seconds, so their edges are blurry, and an edge that is 1 s too early would remove the first moment of the action that follows, the one DA3 and the VLM need in order to see the transition. Better to keep one second of dead footage than to lose the start of a move.

Removed, remainder stitched. The shrunk runs are cut out with `ffmpeg` and the surviving pieces are concatenated into a single continuous video, the “stripped” file the later stages consume. Alongside it a kept-map records that, for example, stitched second 103 corresponds to source second 110, so every timestamp reported downstream can be converted back to the original recording.

On mov-01 this is where the 279 s tail goes: one long run, trimmed by a second at the front, cut out, leaving a 5-minute stitched video from a 10-minute original.

B.4 The “Alternatives” paragraph of Section 2.2, in easy words: what was tried before the four detectors, and why was each dropped?

A plain motion threshold, or `ffmpeg`’s `freezedetect`. The obvious idea: “if nothing moves, it is dead.” It fails in both directions. The most common dead footage in this corpus is not a frozen frame; it is the camera swinging around while the wearer walks to the next spot, or picks the camera up and puts it down. Lots of motion, nothing useful, and a motion threshold keeps all of it. Meanwhile the 2 to 3 s pauses between two actions have no motion, so the threshold cuts them, and those pauses are exactly where one sub-task ends and the next begins. It throws away the boundaries and keeps the junk.

An earlier quantile-based detector. The first version of the current idea set its threshold from percentiles of the same video’s motion, such as “below the 20th percentile is dead”. On a video that is half dead, the 20th percentile lies inside the dead region itself, so the threshold sits in the middle of the noise of the dead stretch and the detector flickers on and off within it. On mov-01 it found 88 s of a roughly 240 s dead region, in five separate pieces. The fix was comparing against the median and the 90th percentile, which sit in the active part, and adding the absolute floors. That detector also ran on a hand-written RANSAC homography, which had its own problems (App. B.5).

Per-frame thresholding without run extraction. Deciding frame by frame and cutting every dead frame individually leaves a video chopped into hundreds of tiny pieces. That is why App. B.3 exists: group into runs, require ≥ 5 s, pad by 1 s.

More expensive checks not needed. The original plan had a sequence of increasingly expensive checks in case the cheap ones failed: comparing image embeddings over time to spot a scene that is not changing, asking the VLM “is anything happening here?” for each suspected run, and using audio loudness. None were built, because the four cheap rules handled all 20 videos.

B.5 How did the RANSAC homography attempt fail, and why is the still detector ORB + MAGSAC instead?

The first camera-motion compensation, a Harris + NCC + RANSAC homography, was written from scratch in numpy: Harris corners [14], matched by 9×9 normalised cross-correlation [15] within ± 14 px, a homography fitted by plain RANSAC [16] (500 four-point samples solved by the normalised DLT [17], 2px threshold), then the same warp-and-subtract residual the still detector uses today. It did the job on calm footage and failed in three ways.

It could not follow a fast camera. The matcher only searched ± 14 px around each corner. On mov-04 the camera moves 18px per frame at the median and 51px at the 90th percentile (ORB later measured 28 / 174px), so 31 % of frame pairs failed outright, the inlier ratio on the rest was 0.51, and compensation removed nothing: residual / raw = 0.98. Worse, a low inlier ratio was being read as “lots of independent motion” when it meant “the estimated transform is unreliable”. mov-04 is also the video with the most walking-and-repositioning dead time, the case a naive motion gate cannot see, so this is exactly where the homography was needed.

It made a static camera worse. On mov-05 the camera barely moves (0.09px median). Warping through a near-identity homography adds bilinear interpolation error to every pixel, so the compensated residual came out 36 % *larger* than the raw frame difference. Compensation has to be skipped when $H \approx I$.

It produced nothing on blank footage, and the thresholds followed it. From ~ 360 s mov-01 is a camera on a featureless surface: Harris finds no corners, every pair fails, and the quantile thresholds computed over the whole video were then set by that dead region, giving the on-off switching described in App. B.4.

Also slow: 160 to 320s per ten-minute video in pure numpy. The replacement keeps the idea and changes the parts: OpenCV ORB (a few hundred features, matched by descriptor rather than a local search, so fast motion is fine) and OpenCV’s `findHomography` with the `USAC_MAGSAC` flag, i.e. MAGSAC++ [13], a robust estimator that does not need a hand-set inlier threshold, roughly $10\times$ faster; absolute floors instead of video quantiles; blank frames handled by the texture and blank gates rather than by the homography; and pairs where no homography can be fitted return no result instead of voting. The 2×2 dead-footage classifier from that experiment (camera motion $\times$ independent motion) was dropped with it; the four detectors of Section 2.2 replaced it.

B.6 What are a “station” and a “station change”, what does “finishes at one station” mean, and what if the person moves quickly?

A station is the spot where a piece of work is done: a workbench, a sink, a shelf, a table. “Finishes at one station” means the person completes what they were doing at that spot and stays there for a few seconds before turning and walking to the next spot. Example from mov-04: the wearer wipes a table (station 1), goes still, turns at about 0:29, walks across the room and starts on the counter (station 2); the cut is placed at that turn. The word is used so that a sub-task is defined by *where* the work happens rather than by what the hands are doing, which is what the camera trajectory can actually measure.

A station change is the moment the person leaves one station for another: they stop working at spot A, turn, walk to spot B and start working there. It is the unit of boundary this pipeline places, and it is what every “cut” in the station table is. Concretely it is a point in time, placed at the walk onset after the turn, at which the template’s conditions all hold: still at A for 5s, a turn $\geq 60^\circ$, a walk $\geq d$ within 10s, and staying $\geq 0.7d$ away for 10s. Two tasks done at the

same spot, such as wiping the counter and then the mirror above it, are not a station change and get no cut from the pose rule; their boundary, if any, comes from the timestamped actions in the description. Across the 17 analysed videos there are 37 station changes, which is why there are 54 clips.

The still-before threshold. In the 5 s before the turn, the camera positions must stay within a 0.5 m spread (90th percentile of distance from their centre). That is not “frozen”, it is “working in one place”: normal reaching, leaning and stepping around a bench stay inside 0.5 m.

If they move quickly and skip that pause, the full template does not fire, and two things happen:

- The turn and walk are still logged. Every turn $\geq 60^\circ$ followed by a walk $\geq d$ that fails the still-before or stays-away condition is reported as a *partial* match, and a walk $\geq d$ with the stay condition met but no qualifying turn as a *no-turn* departure, so the candidate is visible in the station table but not applied as a cut.
- The walk itself is a gap. A person who leaves a station without pausing was already in motion, and that motion usually began at the end of the previous sub-task, so the previous boundary is close by anyway.

In the 17 analysed videos every cut a person would draw by eye passed the still gate, so it was not loosened. For a corpus of hurried workers the parameters to change are the 5 s window and the 0.5 m spread; lowering the window to 3 s is the first thing to try.

B.7 Why two DA3 models, and why is the one-model setting run as well?

The template asks two questions that need different things from the pose model, and only the bigger model can answer the second.

Rotation needs no scale. A turn is measured in degrees. Any structure-from-motion model gives the rotation between two frames exactly, whatever units it uses for distance. So the cheap DA3-LARGE, at about 0.26 s per window, is enough for “did the head sweep 60° , and when was the peak”.

Translation needs metres. The template says “walked at least d , where d is $0.85\times$ the radius of the space used”. That only means something if all positions in the video are in the same unit. DA3-LARGE is scale-free: each 16-frame window is reconstructed with its own arbitrary scale, so a 2 m walk in one window and a 0.5 m step in another can come out as the same number. Chaining such windows gives a trajectory whose shape is right locally but whose distances are meaningless, and a threshold in metres cannot be applied to it. DA3-NESTED-GIANT-LARGE carries a metric head that predicts absolute scale from the scene itself, so every window comes out in metres, the windows can be chained through their shared frames into one consistent trajectory, and d , the 0.5 m still spread and the 3 m clamp all refer to real distances.

Why not the big one for both. It can be, and the pipeline runs that setting too. The metric model’s rotations, fed to the same turn detector and the same template, find the same station changes as the small model on 35 of 37 (the two that swap are threshold cases: a 40° turn against the 60° gate where the small model measured 64° ; still-before spreads of 0.47 m and 0.50 m against the 0.5 m floor). Its angular-speed curve is smoother, so it detects fewer turns (308 against 340 above 60°) and merges neighbouring turns more often. When the cut was anchored at the peak of the turn that mattered: the peak moved by 1 to 5.5 s on a quarter of the cuts. With the walk-onset anchor the cut is read from the trajectory, which both settings share: 27 of

the 35 shared station changes get the same sample in both settings and 29 lie within 1 s. The remaining six are cases where the person is already walking at the peak of the turn, so the walk onset collapses onto the peak and inherits the models’ disagreement about where the peak is, 1.5 to 6.5 s (see App. B.10). Beyond that the settings differ in which station changes they find, the four threshold cases above, and in cost: the one-model setting drops the DA3-LARGE pass, a third of the pose stage. Both settings are reported side by side in Section 3; the two-model setting remains the default because its turn boundaries are cleaner on this corpus.

B.8 How is $d = \text{clip}(k \cdot L, 0.5 \text{ m}, 3 \text{ m})$ computed, and what exactly does the “stays” condition test?

The formula for d . L is the root-mean-square distance of the camera from its average position over the active part of the video. It is one number that says how big the workspace was: a person who spends ten minutes at one bench gets a small L , a person who criss-crosses a room gets a large one. Then $d = 0.85 \times L$, clipped to the range 0.5 to 3 m (Table 5).

Table 5: L and d on three videos.

video	L	$0.85 \times L$	d used
mov-04, room with several stations	1.57 m	1.33 m	1.34 m
mov-03, mostly one bench	0.27 m	0.23 m	0.50 m (floor)
mov-07, large hall	3.87 m	3.29 m	3.00 m (ceiling)

Scaling with L matters because “moved to another station” means something different in a kitchen and in a warehouse. A fixed 1.5 m would miss every station change in mov-03 and fire on every step in mov-07. The 0.5 m floor stops d from collapsing on a video where the trajectory is almost a point, where pose noise alone would then exceed d . The 3 m ceiling stops it from exploding on a video with one long trip away, which would otherwise make every ordinary station change too short to count. $k = 0.85$ rather than 1.0 was set on mov-04, where the visible walk at 0:29 covered 1.42 m against $L = 1.57$ m: the walk to a new station is typically a little shorter than the radius of the whole space.

The “stays” condition. Once the walk is confirmed, that is once the camera has reached $\geq d$ from where it was before the turn, look at the next 10 s. In at least 80 % of those frames the camera must still be $\geq 0.7d$ from the old place. For mov-04, having walked 1.34 m away, it must remain at least 0.94 m away for 8 of the next 10 s.

This rejects a *go-and-come-back*: turn, walk to the shelf, grab a tool, walk back to the same bench. Every other condition matches, but the person is back within 10 s, so the frames near the old place pull the fraction below 0.8 and it is not a cut. The 0.7 factor leaves room to step around the new station without counting as a return; the 0.8 fraction tolerates a couple of seconds of pose jitter or a brief step back. A turn and walk that passes the still and reach conditions but fails this one is reported in the station table as “walked but failed a dwell”, which is the case for the seven such rows on mov-02, where the person keeps shuttling between two spots less than 10 s apart.

B.9 What is a “schema grammar”, and which one did the pipeline use?

One-line definition. A schema grammar is a JSON schema compiled by the inference server into a token-level grammar, so that at every decoding step any token that would make the output invalid under the schema is masked out before sampling; the model can only produce text that matches the schema. In vLLM [6] this is the structured-output path, requested with `response_format = {"type": "json_schema", "json_schema": {...}}`.

The one actually used. The clip schema below is passed unchanged as that grammar in the fallback request and is also the schema the primary path validates against in code.

```
{
  "type": "object",
  "properties": {
    "setting_guess": {"type": "string"},
    "setting_evidence": {"type": "array", "items": {"type": "string"}, "maxItems": 6},
    "camera": {"type": "string"},
    "objects": {"type": "array", "items": {"type": "string"}, "maxItems": 15},
    "actions": {"type": "array", "maxItems": 12, "items": {"type": "object",
      "properties": {"t": {"type": "number"}, "action": {"type": "string"}},
      "required": ["t", "action"], "additionalProperties": false}},
    "others_actions": {"type": "array", "maxItems": 8, "items": {"type": "object",
      "properties": {"t": {"type": "number"}, "action": {"type": "string"}},
      "required": ["t", "action"], "additionalProperties": false}},
    "hands": {"type": "string"},
    "text_or_labels_visible": {"type": "array", "items": {"type": "string"}, "maxItems": 10},
    "one_line_summary": {"type": "string"},
    "visibility": {"type": "string",
      "enum": ["clear", "partly_obscured", "mostly_unclear"]},
    "uncertain_about": {"type": "array", "items": {"type": "string"}, "maxItems": 8},
    "required": ["setting_guess", "setting_evidence", "camera", "objects", "actions",
      "others_actions", "hands", "text_or_labels_visible", "one_line_summary",
      "visibility", "uncertain_about"],
    "additionalProperties": false}
  }
}
```

The global-context request has its own schema of the same kind, with the fields `overall_activity`, `location_guess`, `location_evidence`, `sequence_of_subtasks` (items of start and subtask), `people` and `uncertain_about`.

Small example. Under this schema, once the model has written `{"hands": "both, holding a cloth", "visibility": " "}`, the only tokens it may sample next are the ones that start `clear`, `partly_obscured` or `mostly_unclear`; a value like `"low"` has probability zero, and an extra field such as `"confidence": 0.4` can never appear because `additionalProperties` is false. JSON mode, the primary path, guarantees only that the output parses, so the same answer could come back as `{"hands": "...", "visibility": "low", "confidence": 0.4}` and is caught by the validator in code instead.

Why it is only the fallback. The grammar cannot forbid whitespace between JSON tokens, because whitespace is legal JSON. With thinking on, the model reproducibly drifted into emitting thousands of spaces and newlines between two fields until the 4,000-token budget ran out, returning a truncated record (finish reason `length`). So the primary path is thinking on + JSON mode + code validation, and the grammar is used only in a second attempt with thinking off, where it behaves. Across the 54 clips the fallback was never needed.

B.10 Why is the cut placed at the walk onset, and what does “speed ≥ 0.3 m/s for two consecutive samples” mean?

What the cut should mark. The person has been working in one place, turns, and walks off to the next station. The boundary a viewer would draw is the moment they leave, which is when walking begins, not when the head starts turning and not when the turn is fastest.

Samples and speed. DA3 estimates the camera position twice per second, so the trajectory is a list of positions at 0.0 s, 0.5 s, 1.0 s, 1.5 s and so on; each entry is one *sample*. Speed is the distance from one position to the next divided by half a second, and each speed value is averaged with its neighbour (a two-sample box), so the speed is also a list with one value per half-second, lightly smoothed.

The rule. From the peak of the turn onwards, take the first sample whose speed is at least 0.3 m/s *and* whose next sample is also at least 0.3 m/s. Two neighbouring samples above the threshold means the camera keeps walking pace for one full second; the cut goes at the start of that second. Example, speeds after a turn whose peak is at 2:48.0:

```
time:    2:48.0  2:48.5  2:49.0  2:49.5  2:50.0  2:50.5  2:51.0
speed:   0.1    0.4    0.1    0.2    0.5    0.6    0.7    m/s
```

At 2:48.5 the speed is 0.4, above the threshold, but the next value is 0.1: a single jolt, not walking. At 2:50.0 the speed is 0.5 and the next value is 0.6, so walking has begun, and the cut is placed at 2:50.0. The search runs up to 10s after the turn onset t_0 . If no such pair exists, the first single sample at 0.3 m/s is used; if there is none at all, t_0 . On the corpus the pair was found on 73 of the 74 cut-and-setting rows; the last fallback fired once (mp4-09, two-model setting).

Why two in a row, and why 0.3 m/s. The position estimates jitter by a few centimetres, and one jittery half-second can read as 0.3 m/s on its own, as at 2:48.5 above; asking for a second value above the threshold removes those. 0.3 m/s is well below normal walking speed (about 1 m/s) and well above what pose jitter produces when the person stands still, so the threshold is not sensitive.

The anchors the station script can use, all measured from the same turn event:

anchor	definition	where it lands
onset	t_0 , the start of the rotation event	about 1.5 s before the visible turn, because the event grows backwards through preparatory head motion
peak (anchor in the first runs)	the sample with the highest angular speed inside the turn event	the visible moment of turning away
walk onset (current)	the rule above	0 to 5.5 s after the peak: equal to it when the person walks out of the turn, later when they finish turning, pause, then go
depart	first sample 0.3 m from the pre-turn position	
arrive	first sample at $\geq d$	4 to 10 s after the peak

Three walk-onset rules were compared on all 37 cuts of both settings before choosing: (1) the pure rule above; (2) the same with a guard, keeping the peak when the camera had already reached d before the walk onset (the person walked during the turn and the speed rule latched onto a later movement); (3) the same without the single-sample step, falling back to t_0 . On 67 of 74 cut-and-setting rows the three agree. The guard changed four rows, pulling cuts that had drifted 3 to 5.5 s late back to the peak; the strict fallback changed three rows, moving them 1.5 to 3 s before the peak onto the onset. The pure rule was chosen; the rows where the guard would have applied are visible in the station tables as cuts whose arrival at d precedes the walk onset.

Why it matters for the two settings. The peak comes from whichever model produced the rotation curve, and the two models place it differently. The walk onset comes from the trajectory, which the two settings share, so a station change found by both settings usually gets the same timestamp in both: on mov-01 all seven walk onsets are identical across the settings while the peaks differed on two of them, and over the corpus 27 of the 35 shared cuts fall on the same sample. The exceptions are departures where the person is already walking at the peak: the search starts at the peak, finds walking immediately, and the cut collapses onto the peak, so the two settings differ there by as much as the peaks do (mov-02 3:24.5 against 3:26.5; mov-04 4:08.5

against 4:10.0; six such cuts, 1.5 to 6.5 s apart). Searching from the turn onset instead would remove that dependence but re-introduce the mid-turn cuts the peak anchor was chosen to avoid.

Cost. None: the anchor is chosen after the template has decided, from numbers the script already computes.

References

- [1] Q. Li et al. VITRA: Scalable vision-language-action model pretraining for robotic manipulation with real-life human activity videos. *ICRA*, 2026. arXiv:2510.21571.
- [2] PhysBrain: Human egocentric data as a bridge from vision-language models to physical intelligence. arXiv:2512.16793, 2025.
- [3] H. Lin, S. Chen, J. H. Liew, D. Y. Chen, Z. Li, G. Shi, J. Feng, B. Kang. Depth Anything 3: Recovering the visual space from any views. arXiv:2511.10647, 2025. Models DA3-LARGE-1.1 and DA3NESTED-GIANT-LARGE-1.1, <https://huggingface.co/depth-anything>.
- [4] Qwen Team. Qwen3.8-Max: A new bar for coding and cowork. <https://qwen.ai/blog?id=qwen3.8>, August 2026.
- [5] Qwen Team. Qwen3.8-7B-FP8 model card. <https://huggingface.co/Qwen/Qwen3.8-7B-FP8>, 2026.
- [6] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, I. Stoica. Efficient memory management for large language model serving with PagedAttention. *SOSP*, 2023.
- [7] E. Frantar, R. L. Castro, J. Chen, T. Hoefer, D. Alistarh. MARLIN: Mixed-precision auto-regressive parallel inference on large language models. *PPoPP*, 2025. arXiv:2408.11743.
- [8] Z. Ye, L. Chen, R. Lai, W. Lin, Y. Zhang, S. Wang, T. Chen, B. Kasikci, V. Grover, A. Krishnamurthy, L. Ceze. FlashInfer: Efficient and customizable attention engine for LLM inference serving. *MLSys*, 2025. arXiv:2501.01005.
- [9] A. Gu, T. Dao. Mamba: Linear-time sequence modeling with selective state spaces. *COLM*, 2024. arXiv:2312.00752.
- [10] S. Yang, J. Kautz, A. Hatamizadeh. Gated Delta Networks: Improving Mamba2 with delta rule. *ICLR*, 2025. arXiv:2412.06464.
- [11] G. Farnebäck. Two-frame motion estimation based on polynomial expansion. *Scandinavian Conference on Image Analysis (SCIA)*, LNCS 2749, pp. 363–370, 2003.
- [12] E. Rublee, V. Rabaud, K. Konolige, G. Bradski. ORB: An efficient alternative to SIFT or SURF. *ICCV*, pp. 2564–2571, 2011.
- [13] D. Barath, J. Noskova, M. Ivashechkin, J. Matas. MAGSAC++, a fast, reliable and accurate robust estimator. *CVPR*, pp. 1304–1312, 2020.
- [14] C. Harris, M. Stephens. A combined corner and edge detector. *Alvey Vision Conference*, pp. 147–151, 1988.
- [15] J. P. Lewis. Fast normalized cross-correlation. *Vision Interface*, pp. 120–123, 1995.
- [16] M. A. Fischler, R. C. Bolles. Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981.
- [17] R. Hartley, A. Zisserman. *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.
- [18] N. Otsu. A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1):62–66, 1979.

- [19] A. F. Smeaton, P. Over, A. R. Doherty. Video shot boundary detection: Seven years of TRECVID activity. *Computer Vision and Image Understanding*, 114(4):411–418, 2010.
- [20] G. Bradski. The OpenCV library. *Dr. Dobb’s Journal of Software Tools*, 2000. <https://opencv.org>.
- [21] P. Virtanen et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.
- [22] FFmpeg developers. FFmpeg multimedia framework, with the x264 H.264 encoder. <https://ffmpeg.org>, <https://www.videolan.org/developers/x264.html>.
- [23] Y. A. Farha, J. Gall. MS-TCN: Multi-stage temporal convolutional network for action segmentation. *CVPR*, pp. 3575–3584, 2019.
- [24] F. Yi, H. Wen, T. Jiang. ASFormer: Transformer for action segmentation. *BMVC*, 2021.
- [25] D. Liu, Q. Li, A. Dinh, T. Jiang, M. Shah, C. Xu. Diffusion action segmentation. *ICCV*, 2023.
- [26] Z. Du, X. Wang, G. Zhou, Q. Wang. Fast and unsupervised action boundary detection for action segmentation. *CVPR*, 2022.
- [27] M. S. Sarfraz, N. Murray, V. Sharma, A. Diba, L. Van Gool, R. Stiefelhagen. Temporally-weighted hierarchical clustering for unsupervised action segmentation. *CVPR*, 2021. arXiv:2103.11264.
- [28] M. Z. Shou, S. W. Lei, W. Wang, D. Ghadiyaram, M. Feiszli. Generic event boundary detection: A benchmark for event segmentation. *ICCV*, 2021. arXiv:2101.10511.